NT219 — Cryptography
Week 4: Modern Symmetric Ciphers

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September 29, 2025

Outline

What is Cryptography?

- **Cryptology** = Cryptography + Cryptanalysis
- **Security Properties**:
  - Confidentiality
  - Integrity
  - Authentication
  - Non-repudiation (Accountability)
  - Availability
  - Privacy

Cryptographic Primitives (Grouped)

Encryption (Confidentiality)	Symmetric (AES, ChaCha20)
	Asymmetric (RSA, ECC, CRYSTALS-KYBER)
Integrity & Authentication	Hash functions (SHA-2, SHA-3)
	Message Authentication Codes (HMAC, CMAC)
Authentication & Non-repudiation	Digital Signatures (RSA, ECDSA, Dilithium)
	Digital Certificates (PKI)
Key Establishment	Diffie–Hellman (DH), Elliptic Curve DH (ECDH), Key Encapsulation Mechanisms (KEMs)

Classical cipher algorithms

Substitution techniques

- **Monoalphabetic cipher**: uses a single substitution alphabet (e.g., Caesar, simple substitution).
- **Polyalphabetic cipher**: uses multiple *substitution alphabets* to obscure frequency patterns (e.g., Vigenère).
- **Polygraphic cipher**: substitutes blocks of letters instead of single characters (e.g., Hill cipher, Playfair).

Transposition techniques

- Characters remain the same but their positions are rearranged.
- Examples: Rail fence, Columnar transposition.

Cryptanalysis — Statistical techniques

- **Data & notation.** Total count $N = \sum_i f_i$; empirical frequencies $q_i = f_i/N$; language frequencies p_i (e.g., English).
- **Index of Coincidence (IoC).**

$$\mathrm{IoC} = \frac{\sum_i f_i(f_i-1)}{N(N-1)}, \quad \mathrm{IoC}_{\mathrm{EN}} \approx 0.066,\ \mathrm{IoC}_{\mathrm{rand}} = \tfrac{1}{26} \approx 0.0385.$$

  Heuristic: $\mathrm{IoC} \approx 0.06$ suggests monoalphabetic; ≈ 0.04 suggests long-key polyalphabetic.

- **Friedman key-length estimate (Vigenère).**

$$\hat{k} \;\approx\; \frac{0.0265\,N}{(0.065 - \mathrm{IoC}) + N\,(\mathrm{IoC} - 0.0385)}.$$

- **Caesar/shift recovery (also helpful for simple monoalphabetic).**

$$\chi^2(s) \;=\; \sum_i \frac{(N\,q_i^{(s)} - N\,p_i)^2}{N\,p_i} \;=\; \sum_i \frac{(q_i^{(s)} - p_i)^2}{p_i}, \quad s^\star = \arg\min_s \chi^2(s).$$

  Alternative score: $c(s) = \sum_i q_i^{(s)}\,p_i$, take $\arg\max_s c(s)$.

- **Notes.** Results stabilize for $N \gtrsim 200$; use digram/trigram tests to break ties or confirm key hypotheses.

Cryptanalysis — Polyalphabetic ciphers

- Example: Vigenère cipher.
- Kasiski examination: find repeated substrings, compute GCD of gaps $\rightarrow$ key length.
- Friedman test: IoC-based estimate of key length.
- Once key length known: split into columns and apply frequency analysis.
- Known-plaintext (crib-dragging) speeds up recovery.

Cryptanalysis — Polygraphic ciphers

- Hill cipher:
  - Ciphertext blocks: $C = KP \pmod{26}$.
  - With n plaintext/ciphertext pairs, solve $K = CP^{-1} \pmod{26}$.
  - Requires $\gcd((\det(P) \bmod 26), 26) = 1$.
- Playfair cipher:
  - Digraph substitution.
  - Attacked via digraph frequency and filler patterns.

Cryptanalysis — Transposition ciphers

- Columnar and Rail Fence ciphers.
- Ciphertext is permutation of plaintext characters.
- Attack: guess key length, reconstruct columns/rails.
- Use n-gram scoring to recognise correct plaintext.

Cryptanalysis — Heuristic approaches

- Hill-climbing, simulated annealing, genetic algorithms.
- Useful for monoalphabetic substitution and mixed ciphers.
- Fitness function: n-gram statistics, dictionary match, crib alignment.

Attack workflows

1. Compute statistics (IoC, frequency, n-grams) to classify cipher family.
2. Monoalphabetic $\rightarrow$ substitution solver.
3. Polyalphabetic $\rightarrow$ estimate key length, split, apply frequency.
4. Hill cipher $\rightarrow$ collect n pairs, solve linear system mod 26.
5. Transposition $\rightarrow$ enumerate permutations, use scoring.

Complexity considerations

- Monoalphabetic substitution: 26! keys (infeasible brute force) but solved by statistics.
- Vigenère: reduces to k monoalphabetic problems; feasible for small k.
- Hill cipher: polynomial-time linear algebra, but weak for small n.
- Transposition: factorial search space, pruned by language scoring.

Countermeasures and lessons

- Classical ciphers insecure against modern analysis.
- Use strong, modern primitives (AES-GCM, ChaCha20-Poly1305).
- Random, long keys and no reuse.
- Authenticated encryption to prevent tampering.
- Classical methods useful for education and CTFs.

Practical tips for students

- Start with statistics (IoC, frequency) to classify cipher.
- Use scripts with n-gram scoring and heuristic search.
- Leverage cribs (probable plaintexts) when available.
- For Hill cipher: ensure modular inverses exist before solving.

Stream cipher model (1/2)

- Bit/byte-wise encryption using a keystream.
- Ciphertext $C = P \oplus \text{KS}$ where KS is keystream.
- Keystream derived from secret key and usually an IV/nonce.[1]

[1] Terminology: secret seed / key, IV or nonce, session key (per session in networks).

ChaCha20: Initialization

We define the state as a 4×4 matrix of 32-bit words:

$$\begin{bmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Initialization (round 0):

$$\begin{aligned} a_{00..03} &= (0x61707865,\ 0x3320646e,\ 0x79622d32,\ 0x6b206574), \\ a_{10..13} &= (k_0,\ k_1,\ k_2,\ k_3), \\ a_{20..23} &= (k_4,\ k_5,\ k_6,\ k_7), \\ a_{30..33} &= (\text{counter},\ n_0,\ n_1,\ n_2). \end{aligned}$$

ChaCha20 rounds (1/3): Column round

Column round: apply the quarter-round to the 4 columns:

$$(a_{00}, a_{10}, a_{20}, a_{30}), \quad (a_{01}, a_{11}, a_{21}, a_{31}),$$

$$(a_{02}, a_{12}, a_{22}, a_{32}), \quad (a_{03}, a_{13}, a_{23}, a_{33})$$

ChaCha20 rounds (2/3): Diagonal round

Diagonal round: apply the quarter-round to diagonals:

$$(a_{00}, a_{11}, a_{22}, a_{33}), \quad (a_{01}, a_{12}, a_{23}, a_{30}),$$
$$(a_{02}, a_{13}, a_{20}, a_{31}), \quad (a_{03}, a_{10}, a_{21}, a_{32})$$

Quarter-round function on (a, b, c, d)**:**

$$\begin{aligned} a &= a + b \pmod{2^{32}}, & d &= (d \oplus a) \lll 16, \\ c &= c + d \pmod{2^{32}}, & b &= (b \oplus c) \lll 12, \\ a &= a + b \pmod{2^{32}}, & d &= (d \oplus a) \lll 8, \\ c &= c + d \pmod{2^{32}}, & b &= (b \oplus c) \lll 7 \end{aligned}$$

Total: 20 rounds = 10 double-rounds (Column + Diagonal).

Stream cipher model (3/3)

- Security pitfalls:
  - Keystream reuse (two-time pad)[2] $\Rightarrow C_1 \oplus C_2 = P_1 \oplus P_2$.
  - Poor PRG design[3] $\Rightarrow$ state recovery under statistical biases.
  - Weak or predictable IV/nonce selection[4] $\Rightarrow$ replay or collision attacks.
- Cryptanalysis of stream ciphers:
  - **State recovery attacks**[5]: reconstruct internal state from keystream outputs.
  - **Distinguishing attacks**[6]: detect non-randomness in keystream vs true randomness.
  - **Correlation attacks**[7]: exploit linear relation between keystream bits and secret key bits.
  - **Algebraic attacks**[8]: model keystream generation equations and solve for key/state.
  - **Side-channel attacks**[9]: recover keys from timing, power, or cache leakage.

[2] Sử dụng lại chuỗi khóa, hay "two-time pad", có thể dẫn đến lộ thông tin khi XOR hai ciphertext.

[3] Thiết kế bộ sinh số giả ngẫu nhiên (PRG) kém có thể khiến trạng thái nội bộ bị khôi phục dựa trên các thiên lệch thống kê.

[4] Chọn IV hoặc nonce yếu hoặc dự đoán được có thể dẫn đến tấn công tái phát hoặc va chạm.

[5] Tấn công phục hồi trạng thái: xây dựng lại trạng thái nội bộ từ các output của keystream.

[6] Tấn công phân biệt: phát hiện tính không ngẫu nhiên trong keystream so với chuỗi ngẫu nhiên thật.

[7] Tấn công tương quan: khai thác mối quan hệ tuyến tính giữa các bit keystream và bit khóa bí mật.

[8] Tấn công đại số: mô hình hóa các phương trình sinh keystream và giải để tìm khóa hoặc trạng thái.

[9] Tấn công kênh bên: khôi phục khóa dựa trên thông tin rò rỉ từ thời gian, điện năng hoặc cache.

Block cipher basics

- Map a *block* of plaintext to a *block* of ciphertext (same length).
- Typical block sizes: 64 or 128 bits.
- Shared symmetric key; widely used in network applications.

Stream vs. Block ciphers

- **Stream**[10]: bit/byte granularity, low latency, careful nonce handling.
- **Block**[11]: block granularity, modes of operation (ECB, CBC, CTR, GCM, ...).
- Choice depends on use case (throughput, latency, AEAD needs).

[10] Granularity: mức độ chi tiết xử lý dữ liệu. Ở cipher dạng stream, dữ liệu được xử lý theo từng bit hoặc byte.
[11] Ở cipher dạng block, dữ liệu được xử lý theo từng khối cố định (block).

Substitution tables

Type	**Input granularity**	**Description / Key space**
Monoalphabetic substitution	bit / byte (single symbol)	One-to-one mapping of each symbol
Block substitution (S-box)	n-bit block	Maps n-bit block to n-bit block; key

Table: Comparison of substitution types and their key space

Question

How many possible substitutions exist for an n-bit block?

Answer:

Feistel structure: intuition

- Alternates **substitutions**[12] and **permutations**[13].
- Implements Shannon's **confusion**[14] & **diffusion**[15].
- Popular for many block ciphers, e.g., DES[16], Blowfish[17], CAST-128[18], IDEA (modified Feistel)[19], Twofish[20].

[12] Substitution: thay thế dữ liệu, ví dụ ánh xạ ký tự trong Caesar cipher hoặc khối trong S-box.
[13] Permutation: tráo đổi vị trí dữ liệu, ví dụ tráo thứ tự bit hoặc byte trong cipher block.
[14] Confusion: làm phức tạp mối quan hệ giữa khóa và ciphertext, giảm khả năng suy đoán khóa.
[15] Diffusion: phân tán thông tin plaintext khắp ciphertext, giảm tính lộ dữ liệu.
[16] DES: Data Encryption Standard, cipher khối cổ điển sử dụng Feistel với 16 vòng.
[17] Blowfish: cipher khối sử dụng Feistel với 16 vòng, block 64-bit, khóa biến đổi từ 32–448 bit.
[18] CAST-128: cipher khối Feistel 64-bit, 12–16 vòng, dùng trong PGP và TLS.
[19] IDEA: sử dụng cấu trúc gần giống Feistel, block 64-bit, 8.5 vòng; nổi bật trong PGP.
[20] Twofish: cipher khối 128-bit, Feistel-style, khóa 128/192/256-bit, an toàn, từng là ứng viên AES.

Confusion and diffusion (Shannon)

Diffusion[a]

[a]Diffusion: phân tán thông tin plaintext khắp ciphertext, làm giảm khả năng rút trích thông tin thống kê.

Example: Plaintext: 1010 1100 0110 1001 After diffusion (simplified): 1101 0011 1010 0110

- Một bit thay đổi trong plaintext ảnh hưởng nhiều bit trong ciphertext.
- Thống kê ký tự/bit ban đầu không còn trực tiếp nhận ra (**(as in DES or AES diffusion algorithm)**).

Confusion[a]

[a]Confusion: làm phức tạp mối quan hệ giữa key và ciphertext, giảm khả năng suy đoán khóa.

Example: Ciphertext $= F(\text{plaintext}, K)$

- Thay đổi 1 bit của khóa $K \rightarrow$ nhiều bit ciphertext thay đổi không dự đoán được.
- Giúp bảo vệ khóa khỏi các tấn công phân tích thống kê.

Feistel encryption/decryption (16 rounds)

Setup:

- Block size: 2ℓ bits, split $M = L_0 \| R_0$ with $|L_0| = |R_0| = \ell$.
- Number of rounds: $n = 16$ (example for DES).
- Round keys: $K_1, K_2, \ldots, K_n$.

Encryption (round $i = 1, \ldots, n$):

$$L_i = R_{i-1} \qquad \text{(left becomes previous right)}$$
$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i) \qquad \text{(right = previous left XOR round function)}$$

Ciphertext:

$$C = L_{n+1} \| R_{n+1} \quad \text{with } L_{n+1} = R_n,\ R_{n+1} = L_n$$

Decryption:

- Same structure as encryption.
- Use subkeys in reverse order: $K_n, K_{n-1}, \ldots, K_1$.

Tip for visualization: Draw L/R boxes and arrows per round to see the swap and F-function effect.

Correctness proof (sketch)

By induction, show for decryption state $L'_i = R_{n-i}$ and $R'_i = L_{n-i}$ for $i = 0, \ldots, n$; thus $L'_{n+1} \| R'_{n+1} = L_0 \| R_0 = M$.

Feistel design features

- **Block size**: larger generally $\rightarrow$ better security, slower speed.
- **Key size**: larger $\rightarrow$ better brute-force resistance.
- #**Rounds**: security grows with rounds (single round is inadequate).
- **Round function** F: nonlinearity/complexity improves resistance.
- **Subkey generation**: make recovering subkeys/main key hard.
- **Implementation**: fast software paths, clear specification to aid analysis.

DES overview

- FIPS 46 (1977); dominant until AES (2001).
- 64-bit block; 56-bit key (64 bits with parity).
- 16-round Feistel; same key for encryption/decryption.

DES encryption outline

Setup:

- Input plaintext M (64-bit)
- Initial permutation: $\mathrm{IP}(M) = L_0 \| R_0$ with $|L_0| = |R_0| = 32$
- Subkeys: $K_1, K_2, \ldots, K_{16}$

Encryption rounds ($i = 1, \ldots, 16$):

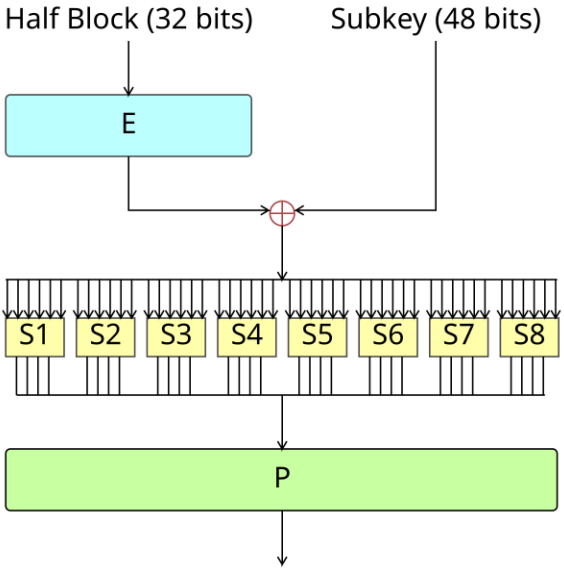
$$L_i = R_{i-1} \qquad \text{(left half becomes previous right)}$$
$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i) \qquad \text{(right = previous left XOR round function)}$$

Output:

- Preoutput: $R_{16} \| L_{16}$ (note the swap)
- Ciphertext: $C = \mathrm{IP}^{-1}(R_{16} \| L_{16})$

Tip: Draw L/R boxes and arrows per round to visualize Feistel structure.

DES round function F



DES round function F

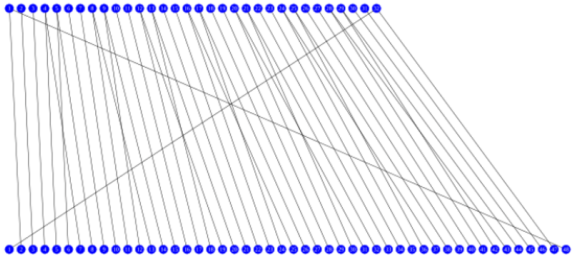
$$F(R_{i-1}, K_i) = P(S(\mathrm{EP}(R_{i-1}) \oplus K_i)).$$

- Expansion $\mathrm{EP} : 32 \to 48$ bits; XOR with K_i.
- Eight S-boxes: $8 \times$ (6-bit $\to$ 4-bit) mappings $\Rightarrow$ 32 bits.
- Final permutation P of 32 bits.

The Feistel Expansion

Expansion function (E) [edit]

E					
32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1



DES S-box Lookup (Example)

Input: 6 bits $b_1 b_2 b_3 b_4 b_5 b_6$

Step 1: Determine row and column

- Row: $b_1 b_6$ (2 bits, 0–3)
- Column: $b_2 b_3 b_4 b_5$ (4 bits, 0–15)

Step 2: Lookup S-box table (S_I for block i=1...8)

- Use the row and column to find the 4-bit output.

Example (S_5):

- Input: $b = 101011$
- Row: $b_1 b_6 = 11_2 = 3$
- Column: $b_2 b_3 b_4 b_5 = 0101_2 = 5$
- Output from S_5: 1001_2 (example)

21

[21] DES S-boxes: 8 tables, each maps 6-bit input → 4-bit output to provide nonlinearity.

DES Lookup table S-box

Input: 011011 –> 0 1101 1
row: 01; column: 1101
Output: $S5_{01,1101} = 1001$

S_5		Middle 4 bits of input															
		0000	**0001**	**0010**	**0011**	**0100**	**0101**	**0110**	**0111**	**1000**	**1001**	**1010**	**1011**	**1100**	**1101**	**1110**	**1111**
Outer bits	**00**	0010	1100	0100	0001	0111	1010	1011	0110	1000	0101	0011	1111	1101	0000	1110	1001
	01	1110	1011	0010	1100	0100	0111	1101	0001	0101	0000	1111	1010	0011	1001	1000	0110
	10	0100	0010	0001	1011	1010	1101	0111	1000	1111	1001	1100	0101	0110	0011	0000	1110
	11	1011	1000	1100	0111	0001	1110	0010	1101	0110	1111	0000	1001	1010	0100	0101	0011

Key schedule (summary)

- Start from $K = k_1 \dots k_{64}$; remove parity bits $\Rightarrow$ 56 bits.
- Initial key permutation to $U_0 \| V_0$ (28 bits each).
- Round-dependent circular left shift: $U_i = \mathrm{ROTL}_{z(i)}(U_{i-1})$, $V_i = \mathrm{ROTL}_{z(i)}(V_{i-1})$.
- Compression permutation to K_i (48 bits). **Left shifts per round $z(i)$:**

Round i	1	2	3	4	5	6	7	8
$z(i)$	1	1	2	2	2	2	2	2
Round i	9	10	11	12	13	14	15	16
$z(i)$	1	2	2	2	2	2	2	1

Is DES still secure?

- Key space: $2^{56} \approx 7.2 \times 10^{16}$ keys.
- Brute-force milestones:
  - 1997: Internet collaborative effort (months).
  - 1998: EFF custom hardware (days).
  - 1999: Combined approach ($\sim$22 hours).
- Conclusion: DES alone is obsolete; use 3DES (legacy) or AES.

DES cryptanalysis: classical results

- **Differential cryptanalysis** (Biham–Shamir, 1993): first published attack on full 16-round DES better than brute force; uses about 2^{47} chosen plaintexts to collect $\approx 2^{36}$ useful pairs; time $\approx 2^{37}$ encryptions.[22]
- **Linear cryptanalysis** (Matsui, 1993/94): known-plaintext attack; needs about 2^{43} known plaintexts to recover a 16-round DES key; first experimental key recovery reported.[23]
- **Davies–Murphy attack**: exploits biased joint distributions of adjacent S-box outputs; improved to about 2^{50} known-plaintexts (Biham–Biryukov, 1997) and to about 2^{45} chosen-plaintexts (Kunz-Jacques & Muller, 2005).[24]

[22] E. Biham and A. Shamir, *Differential Cryptanalysis of the Data Encryption Standard*, Springer, 1993; authors' LaTeX version (pp. 1–2).

[23] M. Matsui, "Linear Cryptanalysis Method for DES Cipher" (abstract).

[24] E. Biham and A. Biryukov, *Journal of Cryptology* (1997); S. Kunz-Jacques and F. Muller, "New Improvements of Davies–Murphy Cryptanalysis," ASIACRYPT 2005.

DES properties & practical implications

- **Weak / semi-weak keys** and **complementation property**: 4 weak keys and 6 semi-weak key pairs; $E_K(P) = C \Rightarrow E_{\overline{K}}(\overline{P}) = \overline{C}$, which can halve exhaustive search under chosen-plaintext.[25]
- **Double DES is broken by meet-in-the-middle**: effective complexity $\approx 2^{57}$ time with $\approx 2^{56}$ space; motivates 3DES instead of 2DES.[26]
- **Practical brute force**: EFF *Deep Crack* (1998) broke DES in $\sim$56 hours; EFF + distributed.net (1999) in $\sim$22 hours.[27]
- **Current status**: DES is obsolete; TDEA (3DES) withdrawn for new use by NIST.[28]

[25] *Data Encryption Standard* (Security and cryptanalysis); D. W. Davies, "Some Regular Properties of the DES," CRYPTO '82.

[26] "Meet-in-the-middle attack" (Double DES) overview.

[27] EFF press releases on DES Challenge II/III; "EFF DES cracker" historical summary.

[28] NIST CSRC announcement, "NIST to Withdraw SP 800-67 Rev. 2" (effective Jan. 1, 2024); SP 800-131A transition guidance.

Principles (I)

- #Rounds must exceed effort of brute-force search under best-known cryptanalysis.
- Round function should satisfy:
  - **SAC** (Strict Avalanche Criterion): each output bit flips with prob. $\frac{1}{2}$ when any single input bit is inverted.
  - **BIC** (Bit Independence Criterion): output bits change independently under single input-bit inversions.

Principles (II)

- Key schedule should maximize difficulty of deducing subkeys and recovering the master key.
- Algorithm should have good avalanche effect overall.
- Favor designs amenable to fast software while remaining analyzable.

Advanced Encryption Standard (teaser)

- 128-bit block cipher with key sizes 128/192/256.
- Substitution–permutation network (not Feistel).
- We will cover AES structure, MixColumns, SubBytes, ShiftRows, and key schedule in Part 2.

References & Reading

- *Applied Cryptography*, *Cryptography and Network Security*, and standard lecture notes for Chapter 4–6 (block/stream ciphers) and Chapter 5 (DES/AES).
- Feistel, H. (1973). *Scientific American* 228(5), 15–23.
- NIST FIPS 46 (DES) and its supplements.